

4. LOGIC

Propositions and Logical Operators

Definition:

A **proposition** is a statement that has a **truth value**; it is either **True** (labelled T) or **False** (labelled F).

For example:

- *The earth is flat.*
- *Paris is the capital of Spain.*
- *29 is a prime number.*

We usually abbreviate such propositions so that, for example, P stands for *the earth is flat*, Q stands for *Paris is the capital of Spain* and R stands for *29 is a prime number*. Note, that P and Q are False, but R is True.

New propositions can be constructed from others by using logical operators such as **not**, **or** and **and** to produce **compound propositions**.

For example:

- (**not** P) is the proposition *the earth is not flat*.
- (P **or** Q) is the proposition *the earth is flat or Paris is the capital of Spain*.
- (P **and** Q) is the proposition *the earth is flat and Paris is the capital of Spain*.
- (P **and** (**not** Q)) is the proposition *the earth is flat and Paris is not the capital of Spain*.

Truth Tables

We want to determine the truth value of compound propositions. In order to do so we need to know the effect of logical operators. These can be neatly summarized using a **truth table**.

Negation

If P denotes an arbitrary proposition, its **negation** is the proposition (**not** P) whose truth value is the opposite of P .

P	(not P)
T	F
F	T

Conjunction

If P and Q are two propositions, their **conjunction** is the proposition (P **and** Q). This is true, only when P and Q are both true. This agrees with the ordinary usage of the word ‘and’ in the English language. The corresponding truth table is given below.

P	Q	(P and Q)
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction

If P and Q are two propositions, their **disjunction** is the proposition (P **or** Q). This is true when at least one of P or Q is true. This agrees with the ordinary usage of the word “or” in the English language. In other words, (P **or** Q) means “either P or Q or both”. The corresponding truth table is given below.

P	Q	(P or Q)
T	T	T
T	F	T
F	T	T
F	F	F

Using the above, we construct truth tables of more complicated compound propositions. Sometimes, intermediary columns are used to build up each of the expressions.

Example 1:

Construct the truth table for each of the following propositions:

(a) **not** (P **and** Q)

(b) P **or** (**not** Q)

Solution:

(a)

P	Q	(P and Q)	not (P and Q)
T	T	T	F
T	F	F	T
F	T	F	T
F	F	F	T

(b)

P	Q	not Q	P or (not Q)
T	T	F	T
T	F	T	T
F	T	F	F
F	F	T	T

Definition:

Two propositions are **logically equivalent** if they have identical truth values, i.e. if the corresponding columns on the truth table are identical.

Example 2:

Determine if the proposition (**not** (P **and** (**not** Q))) is logically equivalent to the proposition ((**not** P) **or** Q).

Solution:

We construct a combined truth table using the required intermediary columns.

P	Q	not P	not Q	(P and (not Q))	(not (P and (not Q)))	((not P) or Q)
T	T	F	F	F	T	T
T	F	F	T	T	F	F
F	T	T	F	F	T	T
F	F	T	T	F	T	T

Since the last two columns are identical, the two propositions are logically equivalent.

Conditional Proposition

There is a further logical operator which results in a **conditional proposition**. An example of such a proposition is *if tomorrow is Thursday then today is Wednesday*. In logic the conditional proposition *if P then Q* (denoted by $P \Rightarrow Q$) is defined to be false whenever the **premise** P is true and the **conclusion** Q is false. Otherwise, it is defined to be true. ($P \Rightarrow Q$) sometimes reads P *implies* Q or P *is sufficient for* Q or Q *is necessary for* P. The truth table for ($P \Rightarrow Q$) is given below:

P	Q	($P \Rightarrow Q$)
T	T	T
T	F	F
F	T	T
F	F	T

Example 3:

Determine if the proposition ($P \Rightarrow Q$) is logically equivalent to the proposition $[(\text{not } Q) \Rightarrow (\text{not } P)]$.

Solution:

P	Q	not P	not Q	($P \Rightarrow Q$)	$[(\text{not } Q) \Rightarrow (\text{not } P)]$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Since the last two columns are identical, the two propositions are logically equivalent.

Definition:

- A **tautology** is a proposition whose truth values are all true.
- A **contradiction** is a proposition whose truth values are all false.

Example 4:

Determine if each of the following propositions is a tautology, a contradiction or neither.

(a) $(\text{not } P) \Rightarrow [Q \text{ or } (\text{not } Q)]$

(b) $(P \text{ and } Q) \text{ and } [(\text{not } P) \text{ or } (\text{not } Q)]$

Solution:

(a)

P	Q	not P	not Q	Q or (not Q)	(not P) $\Rightarrow$ [Q or (not Q)]
T	T	F	F	T	T
T	F	F	T	T	T
F	T	T	F	T	T
F	F	T	T	T	T

Since all the truth values are true, we conclude that the proposition is a tautology.

(b)

For simplicity, let $S = (P \text{ and } Q) \text{ and } [(\text{not } P) \text{ or } (\text{not } Q)]$

P	Q	P and Q	(not P)	(not Q)	(not P) or (not Q)	S
T	T	T	F	F	F	F
T	F	F	F	T	T	F
F	T	F	T	F	T	F
F	F	F	T	T	T	F

Since all the truth values are false, we conclude that the proposition is a contradiction.

We can have compound propositions constructed from three other propositions (instead of two). It's examples like the following that are more likely to appear in the final exam.

Example 5:

Construct the truth table of the proposition: $[P \text{ and } (\text{not } Q)] \text{ or } [\text{not } (Q \text{ or } R)]$

Solution:

For simplicity, let $S = [P \text{ and } (\text{not } Q)] \text{ or } [\text{not } (Q \text{ or } R)]$

P	Q	R	not Q	P and (not Q)	Q or R	not (Q or R)	S
T	T	T	F	F	T	F	F
T	T	F	F	F	T	F	F
T	F	T	T	T	T	F	T
T	F	F	T	T	F	T	T
F	T	T	F	F	T	F	F
F	T	F	F	F	T	F	F
F	F	T	T	F	T	F	F
F	F	F	T	F	F	T	T

Example 6:

Construct the truth table of the proposition: $[C \Rightarrow (\text{not } B)] \Rightarrow (A \text{ or } B)$

Solution:

A	B	C	not B	A or B	$C \Rightarrow (\text{not } B)$	$[C \Rightarrow (\text{not } B)] \Rightarrow (A \text{ or } B)$
T	T	T	F	T	F	T
T	T	F	F	T	T	T
T	F	T	T	T	T	T
T	F	F	T	T	T	T
F	T	T	F	T	F	T
F	T	F	F	T	T	T
F	F	T	T	F	T	F
F	F	F	T	F	T	F

Example 7:

Construct the truth table of the proposition: $[(\text{not } Q) \text{ and } (P \Rightarrow Q)] \Rightarrow (\text{not } R)$

Solution:

For simplicity, let $S = [(\text{not } Q) \text{ and } (P \Rightarrow Q)] \Rightarrow (\text{not } R)$.

P	Q	R	not Q	not R	$P \Rightarrow Q$	$[(\text{not } Q) \text{ and } (P \Rightarrow Q)]$	S
T	T	T	F	F	T	F	T
T	T	F	F	T	T	F	T
T	F	T	T	F	F	F	T
T	F	F	T	T	F	F	T
F	T	T	F	F	T	F	T
F	T	F	F	T	T	F	T
F	F	T	T	F	T	T	F
F	F	F	T	T	T	T	T